

ONE SMALL STEP FOR AI, One Giant Step for Robotics



Janani Gopalakrishnan Wikram is a technically-qualified freelance writer, editor and hands-on mom based in Chennai

There have been immense and innumerable developments in robotics in recent times—some significant, some not so. Right from form factor and flexibility to motion, sensing and interaction, every aspect of robotics has brought them closer to humans. Robots are now assisting in healthcare centres, schools, hospitals, industries, war fronts, rescue centres, homes and almost everywhere else. We must acknowledge that this has come about not merely due to mechanical developments, but mainly due to the increasing intelligence, or so-called smartness, of robots.

Smartness is a subjective thing. But in the context of robots, we can say that smartness is a robot's ability to autonomously or semi-autonomously perceive and understand its environment, learn to do things and respond to situations, and mingle safely with humans. This means that it should be able to think and even decide to a certain extent, like we do.

Let us take you through some assorted developments from around the world that are empowering robots with these capabilities.

Robot asks questions to clarify confusing instructions (Image courtesy: Brown University)



At a glance

- Deep learning is expected to help robots learn, think and decide like humans.
- Researchers are attempting to run deep learning on small, low-power chips so robots can be independently smart without relying on the Cloud.
- Smartness together with social presence makes robots suitable for tasks like care-giving and education.
- Interaction with robots is getting more natural—through gestures, questions and thoughts.
- Neuromorphic and analogue computing are expected to make a comeback in the fields of AI and robotics.
- Robots are getting closer to thinking like humans. So what is the possibility of revolt?

Understanding by asking questions

When somebody asks us to fetch something, and we do not really understand which object to fetch or where it is, what do we do? We usually ask questions to zero in on the right object. This is exactly what researchers at Brown University, USA, want their robots to be able to do.

Stefanie Tellex of Humans to Robots Lab of Brown University is using a social approach to improve the accuracy with which robots follow human instructions. The system, called FETCH-POMDP, enables the robots to model their own confusion and solve it by asking relevant questions.

The system can understand gestures, associate these with what the human being is saying and use this to understand instructions better. Only when it is unable to do so does it start asking questions. For example, if you signal at the sink and ask the robot to fetch a bowl, and if there is only one bowl in the sink, it will fetch it without asking any questions. But if it finds more than one bowl there, it might ask questions about the size or colour of the bowl. When testing the system, the researchers expected the robot to respond faster when it had no questions to